APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: **Carbon and Its Versatile Compounds**

SUBJECT: Chemistry (CHM101)

CLASS & SECTION: Class 10 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: **95 / 100 (Grade A)**

TEACHER DIRECTIVE & PROBLEM FOCUS:
Explain tetravalency, catenation, allotropes, homologous series, and soap micelle cleansing mechanisms.

1. Executive Overview & Problem Context

Carbon (Atomic Number 6, Electronic Configuration 2,4) occupies a preeminent position in chemistry. Over ten million distinct carbon compounds are known—exceeding the total compounds formed by all other periodic elements combined—attributable to carbon's unique tetravalent hybridization and unparalleled catenation capabilities.

2. Core Analysis, Solutions & Systematic Working

- **Tetravalency & Catenation: The Twin Drivers:**

  Carbon possesses 4 valence electrons. Achieving a stable octet via ionic electron transfer (forming C t z or C t {) is energetically prohibited due to ionization enthalpy and nuclear charge constraints. Hence, carbon forms four stable covalent electron-pair bonds. Catenation represents carbon's innate ability to link covalently with other carbon atoms into extensive linear chains, branched networks, and closed rings.

- **Allotropes: Diamond, Graphite & Fullerenes:**

  Diamond features an sp^3 tetrahedral lattice where every carbon bonds to four others with a bond length of 1.54 Å, producing extreme hardness and zero electrical conductivity. Graphite features sp^2 hexagonal planar sheets bonded by weak Van der Waals forces with delocalized pi electrons, granting high electrical conductivity and lubricating cleavage. Buckminsterfullerene (C-60) forms a spherical soccer-ball truncated icosahedron.

- **Homologous Series & Functional Groups:**

  A family of organic compounds sharing identical functional groups, similar chemical reactivity, and conforming to a general molecular formula, where adjacent members differ by a -CH ,- methylene unit (14 unified mass units): Alkanes (C ™H , ™ Š ,), Alkenes (C ™H , ™), A (C ™H , ™ ‹ ,), Alcohols (-OH), Aldehydes (-CHO), Carboxylic Acids (-COOH).

- **Cleansing Action of Soaps: Micelle Formation:**

  Soap molecules are sodium/potassium salts of long-chain fatty acids (e.g. Sodium Stearate: C • ‡H ƒ …COO {Na z). They exhibit dua amphiphilic nature: a hydrophobic non-polar hydrocarbon tail that dissolves in oily dirt, and a hydrophilic ionic carboxylate head facing aqueous surroundings. In water, they self-assemble into spherical micelles, suspending oily grease in a stable emulsion for rinsing.

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3. Detailed Data Matrix & Systematic Findings

Compound Class	General Formula	Sample Member	Functional Group	Industrial Application
Alkane	C ™H , ™ Š ,	Methane (CH „)	Single C-C bond	CNG & Domestic LPG fuel gas
Alkene	C ™H , ™	Ethene (C ,H „)	Double C=C bond	Polyethylene plastic synthesis
Alcohol	C ™H , ™ Š •OH	Ethanol (C ,H …OH)	Hydroxyl (-OH)	Medical antiseptic & green biofuel
Carboxylic Acid	C ™H , ™ Š •COOH	Ethanoic Acid (CH ƒCOOH)	Carboxyl (-COOH)	Food vinegar & ester perfume synthesis
Ester	R-COO-R'	Ethyl Ethanoate	Ester linkage	Fragrances, flavorings & solvents

4. Student Conclusions & Reflective Synthesis

Organic carbon chemistry forms the physical basis of biochemistry, petrochemical fuels, synthetic pharmaceuticals, and polymers that define modern human technology and living cellular existence.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Analytical Depth & Research Methodology: 28 / 30

Problem Solving & Structured Working: 18 / 20

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding organic chemical depth! Molecular orbital diagrams, isomer structures, and micelle mechanisms are thoroughly detailed."

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